

Open Pandora's Box

Case 1

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Case 2: A 2-month-old boy previously healthy



- Presented with **acute high-grade fever without localizing signs**
- Physical examination at OPD
 - Vital signs: **BT 38.4-39 °C PR 150-180 bpm RR 50 /mins O2sat 99%**
 - GA: A male infant **look drowsiness**
 - HEENT: **AF 2*2 cm tense and bulging, mild dry lip**, PF fingertip, no pale conjunctivae, anicteric sclerae, no discharge per eye, no discharge per nose, no discharge per ear neither oral ulcer nor rash
 - Otolaryngology: seen both side intact tympanic membrane no bulging
 - RS: no tachypnea no retraction clear and equal breath sound both lung
 - CVS: regular, **tachycardia, no murmur**
 - GI: no distention normoactive bowel sound, no discharge per umbilicus
 - Neuro: **drowsiness**, active equal movement both side, Brudzinski negative
 - Extremities: **mottling skin, cold extremities**

What investigations would be appropriate to perform in this case? *kahoot*

- **A: CBC**
- **B: CBC and hemoculture**
- **C: CBC and UA**
- **D: CBC, UA, hemoculture, lumbar puncture**

Case 2: A 2-month-old boy presented with acute high grade fever



Lab investigations:

- **CBC: Hb 10.4 mg/dl, Hct 29.8%, WBC 13,430 cells/cu.mm, neutrophil 65.2%, eosinophil 1.8%, lymphocyte 25.5% monocyte 7.1% platelets 422,000 cells/cu.mm**
- **UA: WBC 1-2 /HPF, RBC 0-1 /HPF, squamous epithelium 0-1 /HPF, Nitrite neg**
- **Lumbar puncture: WBC 0 cells/cu.mm RBC 0 cells/cu.mm (inadequate specimen) protein 27 mg/dl Glucose 75 mg/dl (CSF/serum ratio 0.6)**
- **Hemoculture: pending**



Admission
date



Hemoculture alarm:
Gram-negative bacilli

DOA 1

Impression:

Late-onset neonatal sepsis with suspected bacterial meningitis

Management:

- NSS 15 cc/kg drip in 15 mins then 5%DN/5 IV MT rate 30 ml/hr
- Empiric ATB : ampicillin IV + cefotaxime IV (meningitis dose)

Cefotaxime 300 MKDay

Ampicillin 300 MKDay

Which organisms do you think are responsible for this infection? *kahoot*

- *A. Nontyphoidal Salmonella spp.*
- *B. Escherichia coli*
- *C. Klebsella pneumoniae*
- *D. Pseudomonas auruginosa*

Admission
date



Hemoculture alarm:
Gram-negative bacilli

DOA 1

Acute high-grade fever
without localizing signs
PE: tenses fontanelle

Impression: LONS
Management : Admit,
septic work up & LP

Lab	L101823 Bacteria Hemoculture		
Txn	923025029 - 2 : 1/1	Hn	4281847
ชื่อ	ณัฐพงษ์ หนูไผ่ ช 2 ต		
สิทธิ	A1\$	Room	50-E02
แพทย์ผู้ส่ง	20113	17-03-2568 17:34	
เวลาเก็บ specimen	17-03-2568 : 17:34		
ผู้รับ	ฉัตรชัย คำสุข	17-03-2568 17:54	
ผู้รายงาน	ปริญกรณ์ โนนแก้ว	20-03-2568 15:47	

Request

Bottle1#
(มีเชื้อซ้ำ)

Comment

Preliminary report : gram negative bacilli

Salmonella enterica subsp. enterica (nontyphoidal) Group E

#	Qty	Code	Organism
1		BNSLMN	Salmonella enterica subsp. enterica

#	Code	Antibiotic	MIC	Result
1	AU	Amoxicillin-clavulanic acid	<=4	S
2	FEP	Cefepime	<=1	S
3	CX	Cefotaxime	<=1	S
4	CZ	Ceftazidime	<=1	S
5	TX	Ceftriaxone	<=0.5	S
6	CI	Ciprofloxacin	<=0.06	S
7	DOR	Doripenem	<=0.5	S
8	ETP	Ertapenem	<=0.5	S
9	IP	Imipenem	<=0.5	S
10	LX	Levofloxacin	<=0.06	S
11	ME	Meropenem	<=0.5	S
12	TZ	Piperacillin-tazobactam	<=8	S
13	ST	Trimethoprim-sulfamethoxazole	<=1	S

Cefotaxime 300 MKDay

Ampicillin 300 MKDay



CSF Culture :
Salmonella enterica



Admission
date

Hemoculture alarm:
Salmonella enterica

DOA 1

Acute high-grade fever
without localizing signs
PE: tenses fontanelle

Impression: LONS
Management : Admit,
septic work up & LP

Lab	L101822 Bacteria Routine Aerobic culture		
Txn	1256442 - 4 : 1/1	Hn	4281847
ชื่อ	ผิรุพงษ์ หนูไข่		
ลิส	ipd	Room	PD0317
แพทย์ผู้สั่ง	8574	17-03-2568 23:32	
เวลาเก็บ specimen	17-03-2568 : 23:31		
ผู้รับ	พินิตดา กองสง	17-03-2568 23:51	
ผู้รายงาน	วรรณพร แดงขาวเขียว	20-03-2568 15:48	

#	Qty	Code	Organism
1	Rare	BNSLMN	Salmonella enterica subsp. enterica

#	Code	Antibiotic	MIC	Result
1	AU	Amoxicillin-clavulanic acid		
2	FEP	Cefepime	<=1	S
3	CX	Cefotaxime	<=1	S
4	CZ	Ceftazidime	<=1	S
5	TX	Ceftriaxone	<=0.5	S
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11	ME	Meropenem	<=0.5	S
12	TZ	Piperacillin-tazobactam	<=8	S
13	ST	Trimethoprim-sulfamethoxazole	<=1	S

Request

CSF
@ Comment:-

Comment

Salmonella enterica subsp. enterica (nontyphoidal) Group E

Cefotaxime 300 MKDay

Ampicillin 300 MKDay

Admission
date



Hemoculture & CSF C/S:
Salmonella enterica

DOA 1

Impression: *Salmonella enterica*
septicemia with meningitis

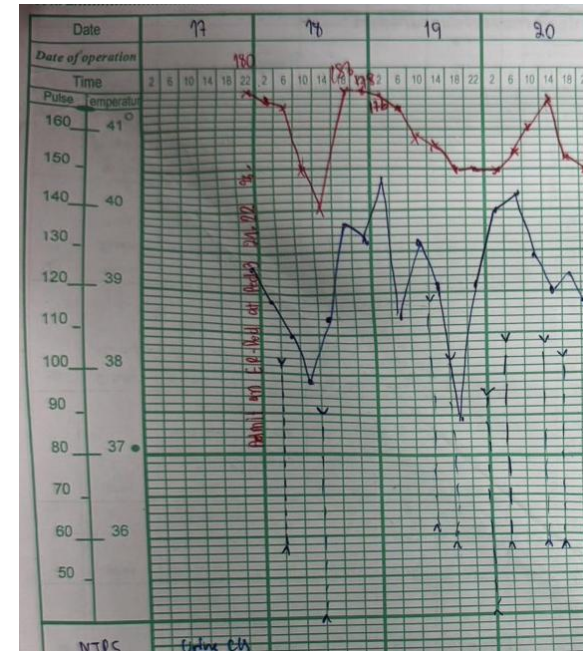
DOA 3

Persistent fever

DDx:

1. Meningitis with complication
2. Drug-resistant organisms

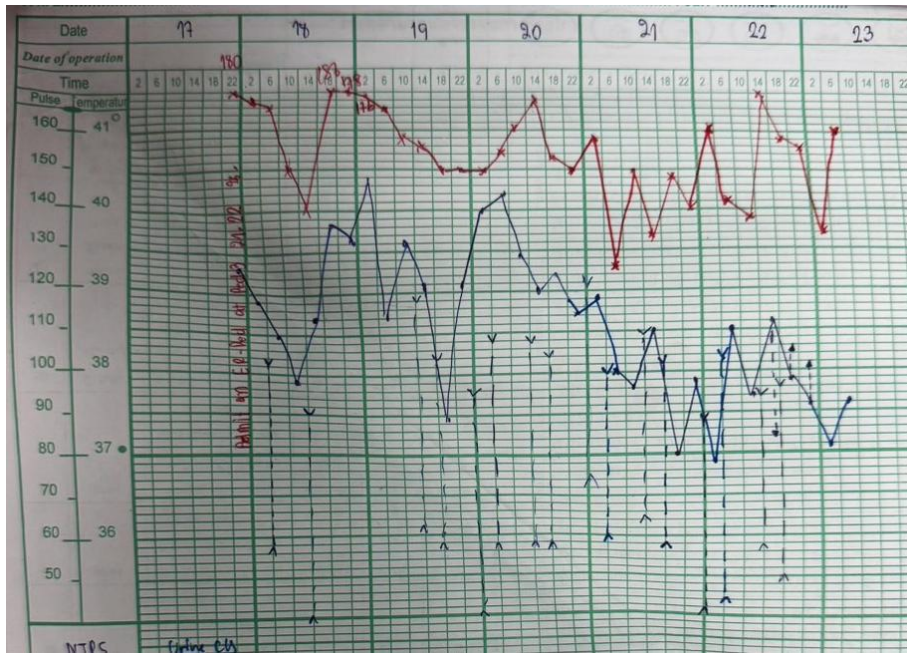
Management: request brain
imaging, repeat LP, step antibiotic



Cefotaxime 300 MKDay

Ampicillin 300 MKDay

Meropenem 120 MKDay



DOA 4

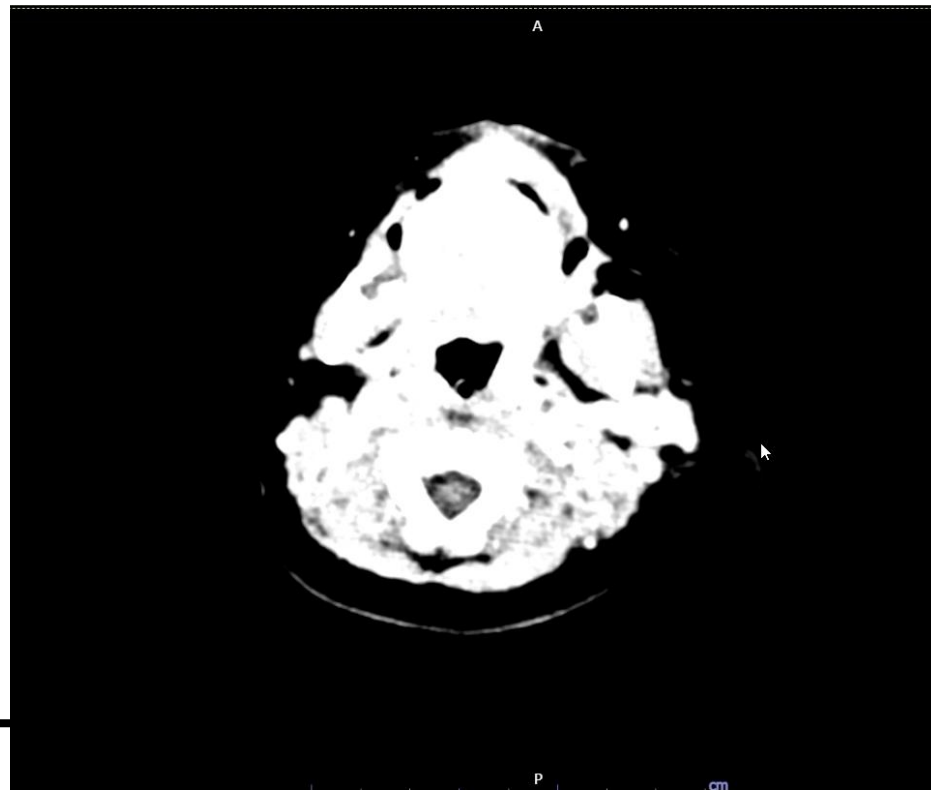
Repeat LP

CSF profile: **WBC 101 cells/cu.mm (PMN 50%, MN 50%) RBC 2**

cells/cu.mm, protein 60 mg/dl, Glucose 41 mg/dl (CSF/serum ratio 0.4)

DOA 7

CT brain with contrast



RESULTS:

- Crescentic markedly hypodense lesions along bilateral fronto-temporo-parietal convexities, measuring up to 1.5 cm in maximal thickness along right frontal convexity, probably subdural collection**

IMPRESSION:

Subdural collection along bilateral FTP convexities

Meropenem 120 MKDay

Ciprofloxacin 30 MKDay

Case 2: A 2-month-old boy previously healthy

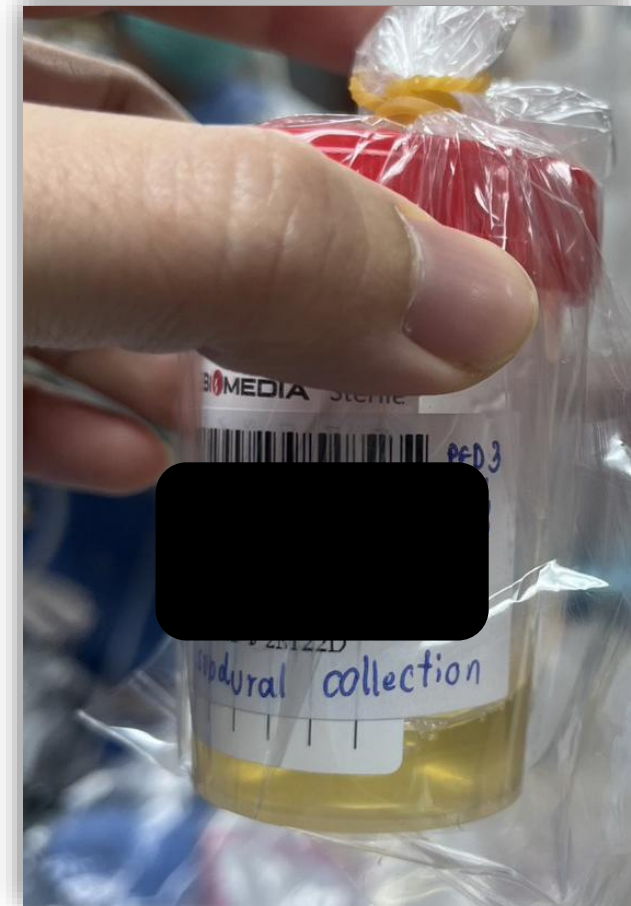


Diagnosis:

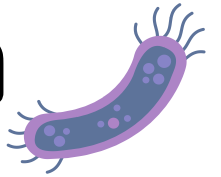
***Salmonella enterica* septicemia with meningitis with subdural collection**

Management

- Consulted Neurosurgery
- Set OR emergency for bilateral burr hole surgery to drain the fluid
- Findings: Serous fluid, total of 53 mL (Left: 3 mL; Right: 50 mL)



Non typhoidal *Salmonella* infection



- One of the leading bacterial causes of diarrhea, causing **≈150 million illnesses** and **≈60,000 deaths globally each year**
- **Most common culture-confirmed foodborne illness** in the United States
- The highest incidence in **children <5 years of age**
- **Young children had the highest incidence rate of NTS BSI, with infants <12 months comprising 22% of all cases**

Transmission



Reservoirs

Animal reservoirs include both domestic and wild animals, including birds, mammals, amphibians, and reptiles



Food vehicles of transmission

Seeded vegetables, poultry, beef, pork, eggs, and dairy products, peanut butter, frozen pot pie, powder infant formula, cereal, bakery products



Other mode of transmission

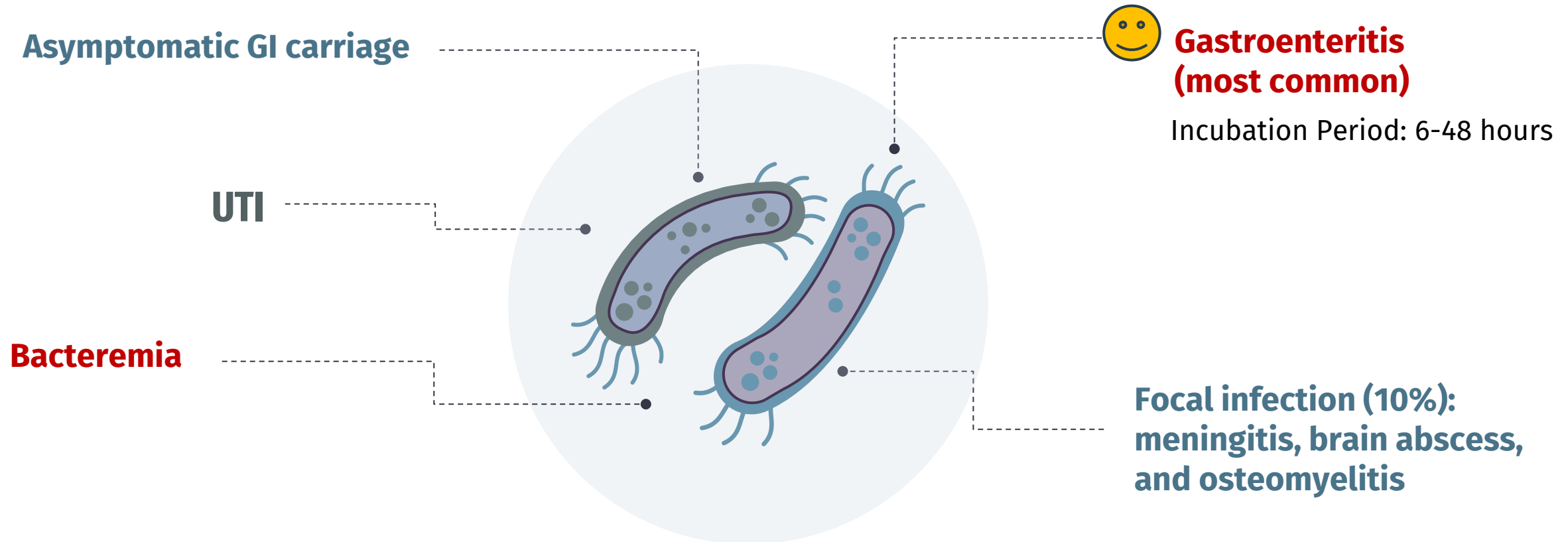
Ingestion **contaminated water or closed contact with infected animals**, poultry (chicken, duck), reptiles, amphibian (pet turtle, iguana, gecko, lizard, snake, frog, salamanders) and rodents (hamster, mice)

Risk of transmission persists as long as infected person sheds NTS organism

- **12 weeks after infection with most common NTS serovars**
- 45% of children younger than 5 shed organism
- ABO prolong shedding
- 1% of adults shed NTS > 1 year

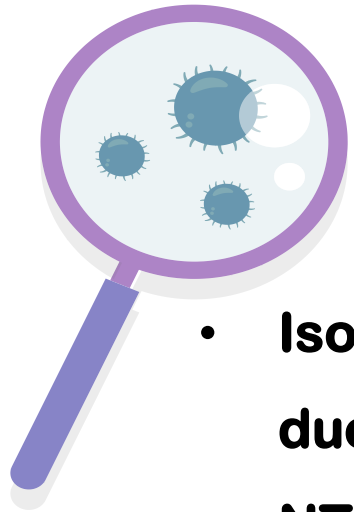
Clinical Manifestations

Non typhoidal salmonella infection



Serovars Dublin, Choleraesuis more likely to result in invasive disease

Serovars most frequent isolate from blood, CSF: Typhimurium, Enteritidis, 1:4,[5],12:i:-



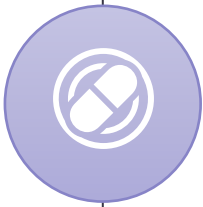
Diagnosis

- Isolation of salmonella from cultures of stool, blood, urine, bile (including duodenal fluid containing bile)
- NTS gastroenteritis: stool culture or molecular testing

Blood cultures should be considered for patients at risk for severe disease (age < 3months, immunocompromised, hemolytic anemia)

Treatment

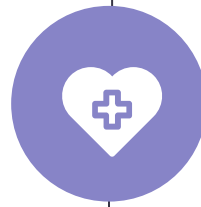
Nontyphoidal Salmonella



Antibiotics Not Indicated

patients with either asymptomatic or uncomplicated gastroenteritis

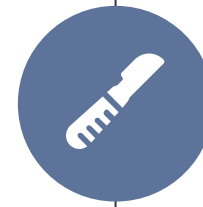
- Not shorten duration of diarrheal disease
- Prolong duration fecal shedding
- Increase symptomatic relapse rate



Antibiotic is recommended

Gastroenteritis in people **at increase risk of invasive disease**

- Infants younger than 3 months
- People with chronic GI disease
- Neoplasm
- Hemoglobinopathies
- HIV infection
- Immunocompromise



Antibiotics

- No evidence of disseminated disease -> OPD case-> oral azithromycin
- Susceptibility test results available-> ampicillin/amoxicillin/bactrim consider for susceptible strain
- Fluoroquinolones (alternative)

Treatment



Bacteremia

Evaluation for disseminated disease (meningitis, osteoarticular, endocarditis)

Antibiotics?

Initial therapy

IV ceftriaxone



Duration

Depend on patient age and other host factor, clinical response

Switch to oral

Transition to oral azithromycin or fluoroquinolone **to complete 7-10 days course**

After blood culture has cleared and focal disease has been excluded



Blood culture

- Repeated until negative

Extended duration

- Meningitis: 4 weeks
- Osteomyelitis: 4-6 weeks



Salmonella meningitis

- **Meningitis are common complications in infants younger than 3 months, 50% to 75% of occurs in the first year of life**
- **A well-appearing infant with Salmonella gastroenteritis may be bacteremic**
- **Malaria has been found to be a risk factor for invasive nontyphoidal Salmonella infections in children**
- **Compared with children with gastroenteritis alone, bacteremic children appear to have a longer duration of symptoms, a less severe clinical appearance, and fewer signs of dehydration**
- **This gradual presentation with less dehydration and fewer toxic effects may lead to premature discharges from emergency departments**



Salmonella meningitis

Treatment

- **Thai CPG of Acute Meningoencephalitis: Cefotaxime/ceftriazone + ciprofloxacin**
- **AAP: recommends that treatment for Salmonella meningitis with cefotaxime or ceftriaxone should continue often for 4 weeks or more**

Elizabeth H. Price (2000); suggest that consideration should be given to antibiotic therapy with a combination of ciprofloxacin and ceftriaxone

- have been used in combination with success for prolonged periods to treat cerebral abscesses caused by Salmonella spp
- Suggest that both agents should be administered for a minimum of 3 weeks and that this should be timed from the first sterile CSF (The specimen should be taken no earlier than day 4 of therapy, as cultures from patients with Salmonella meningitis sometimes remain positive for prolonged periods)